

Multiple Choice (10 marks)

1. Which situation is an example of an inherited trait?
 - (a) lions preying on zebras
 - (b) monkeys using twigs to get food
 - (c) birds following migratory patterns
 - (d) bears opening coolers at campsites
2. What do the Milky Way Galaxy and other galaxies in the universe have in common?
 - (a) They are similar in shape.
 - (b) They rotate in the same direction.
 - (c) They contain the same number of stars.
 - (d) They have similar elements.
3. Most electrical wires are covered with plastic or rubber. The wires are covered with plastic or rubber because those materials
 - (a) are conductors of electricity
 - (b) make complete electric circuits
 - (c) are not conductors of electricity
 - (d) make the electricity move quickly
4. The nearest star to the Sun is described as being 4.3 light-years away. Which statement explains why the light-year unit of measure is used in this description?
 - (a) The distance is extremely large.
 - (b) The distance is a rough estimate.
 - (c) Only light can travel between stars.
 - (d) Light signals were timed reflecting from the star.
5. As the human population grows, the demand for natural resources increases. Which activity wastes a natural resource?
 - (a) turning off lights
 - (b) using public transportation
 - (c) throwing away aluminum cans
 - (d) opening windows to cool a house

True / False (10 marks)

Answer True or False

6. True or False: Recycling centers can separate aluminum and steel cans by sorting them based on their size.
7. True or False: A plastic spoon is made from synthetic materials that are not grown in nature.
8. True or False: Fossil fuels contain chemical energy that can be released through combustion.
9. True or False: The growth stages of a plant progress from seed to young plant and then to mature plant.
10. True or False: Power is measured in watts, which is a unit different from joules.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the phenomenon Alex observes as he watches the sun set behind the neighbor's house, explaining the scientific principles of light, shadow, and the Earth's rotation involved in this event.
12. Discuss the reasons why the output energy of a machine is always less than the input energy, considering factors like friction, energy transformation, and efficiency.

End of Paper